



Project Detail – Modern Spinning Mill with 30,000 Spindles

February 2023



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Executive Summary

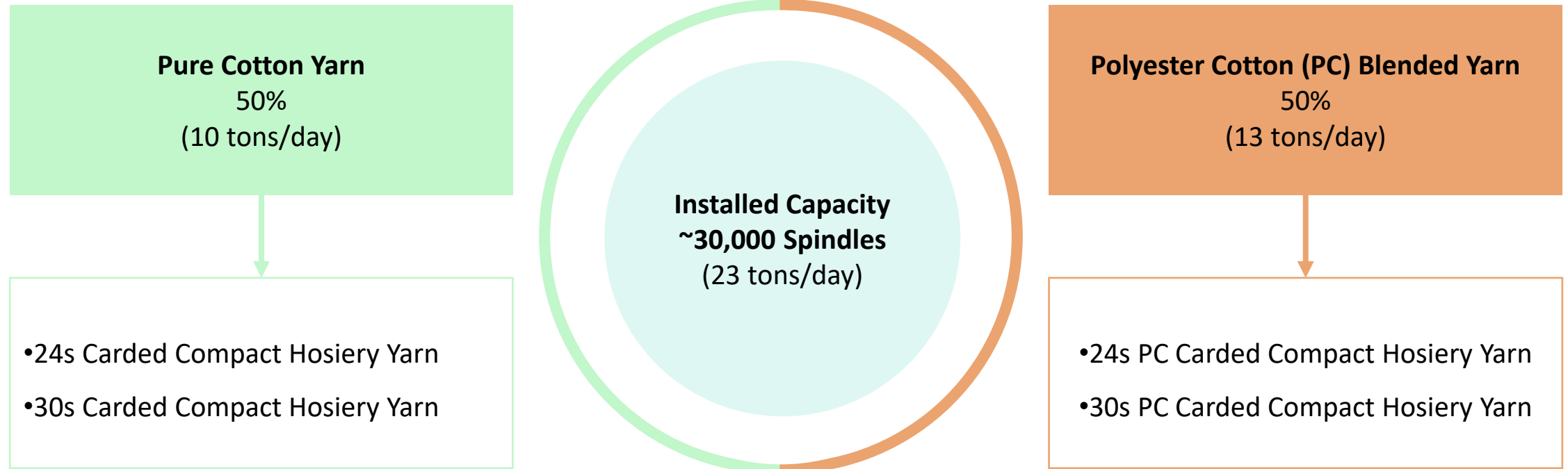
- The proposed spinning mill is a state-of-the-art facility with an *installed capacity of 30,000 spindles (23 tons/day)*.
- The spinning mill will be equipped with the necessary machinery to manufacture both *pure cotton yarn (10 tons/day)* and *polyester cotton blended (PC) yarn (13 tons/day)*.
- The estimated project cost for setting up this modern spinning mill will be *approximately US\$ 27 Mn*.
- Machines with latest technology and automation systems like *bobbin transport system, individual spindle monitoring system, and cone transport system*, have been considered for the mill. These will ensure *low cost of manufacturing, high yarn realization, consistent and sustainable quality of production, and a comfortable working environment for the employees*.
- *Higher inventory and lack of local technicians & skilled manpower* are the key risks, however these can be overcome with an effective mitigation strategy.



Project Snapshot



Proposed Capacity and Product Mix



Project Cost

Values in USD Million

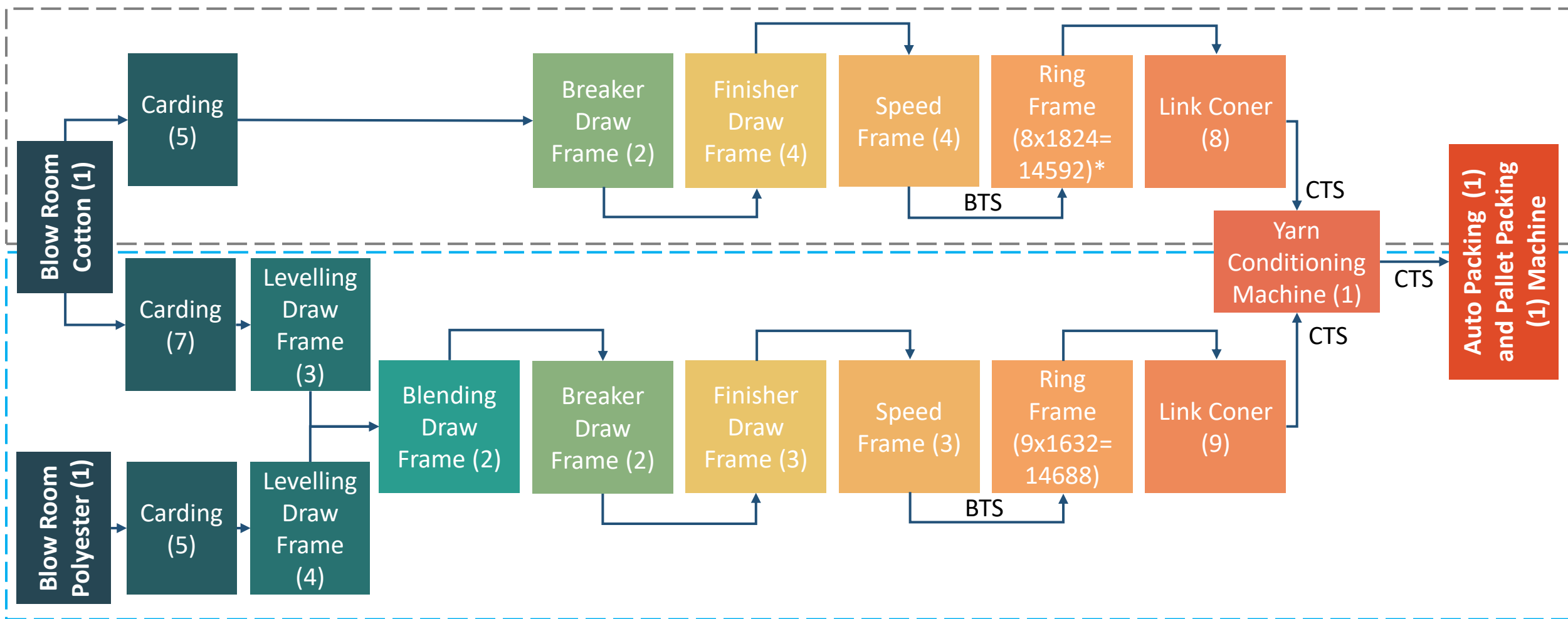
Project Cost	Value	Share	Remark
Land	-		Considering land is already available
Building & Site Development	6.59	25%	Approximately 15,000 sqm of built up area
Plant & Machinery	11.92	45%	State-of-the-art European and Japanese machinery
Miscellaneous Fixed Asset (MFA)	4.26	16%	MFA includes spinning & card room accessories, electrical, utility maintenance and lab equipment, ERP software, etc.
Preliminary and Pre-operative Expenses	1.19	4%	Expenses incurred before the implementation phase like company establishment expenses, interest during construction, travelling expenses, etc.
Provision for Contingencies	2.28	9%	Provision for possible currency fluctuations and change in price while implementation
Margin Money for Working Capital – 1 st Year	0.51	2%	
Project Cost	26.74		

Means of Finance

Values in USD Million

Particular	Share	Value
Equity	10%	2.67
Term Loan	20%	5.35
Grant	70%	18.72
Total		26.74

Proposed Spinning Process



(x) = No. of Machines

*No. of Machines x No. of Spindles = Total Spindles

BTS – Bobbin Transport System

CTS – Cone Transport System

Cotton Yarn (10 tpd)

PC Yarn (13 tpd)

Technological Consideration for State-of-the-Art Spinning Mill

Technological Consideration

- Spinning machines equipped with advanced technology from European and Japanese suppliers
- Automation
 - Bobbin Transport System
 - Individual Spindle Monitoring
 - Cone Transport System
- Cutting-edge Laboratory Equipment
- Solar Panels for Electricity Generation

Benefits of Deploying Latest Technology



- Low Cost of Manufacturing



- Better Yarn Realization (less waste)



- Better and Consistent Quality of Manufacturing



- Comfortable Working Environment



Overview of Spinning Process



Blowroom



- Feed Material – **Fibers (cotton, polyester), re-usable waste**
- Machine Components – **Bale opener, foreign matter separator, cleaners, openers**
- Operations – **Opening, cleaning, mixing, dust removal and uniform feeding to carding**

Carding



- Feed Material – **Cleaned fiber flocks from blowroom via chute feed**
- Machine Components – **Licker-in, cylinder, flats, doffer**
- Operations – **Fiber individualization, elimination of short fiber & dust, disentangling of neps, fiber orientation**

Draw Frame



Machine	Levelling Draw Frame	Blending Draw Frame	Breaker Draw Frame*	Finisher Draw Frame
Feed Material	Carded sliver	Levelled sliver	Blended sliver	Breaker draw frame sliver
Machine Components	Drafting arrangement, creel, coiler, auto leveller			
Operation	Equalizing, parallelizing	Blending	Equalizing, parallelizing	Equalizing, parallelizing

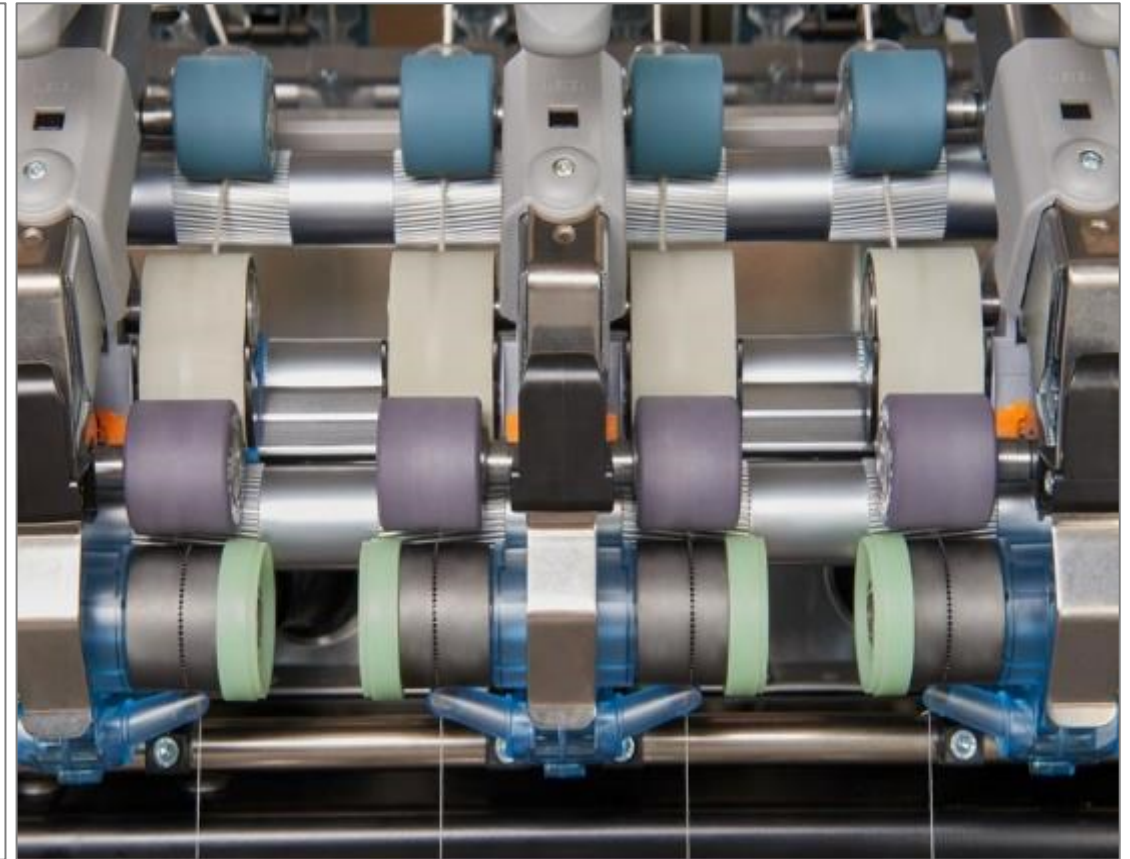
*In case of cotton yarn, the carded sliver will be fed directly to breaker drawframe

Speed Frame



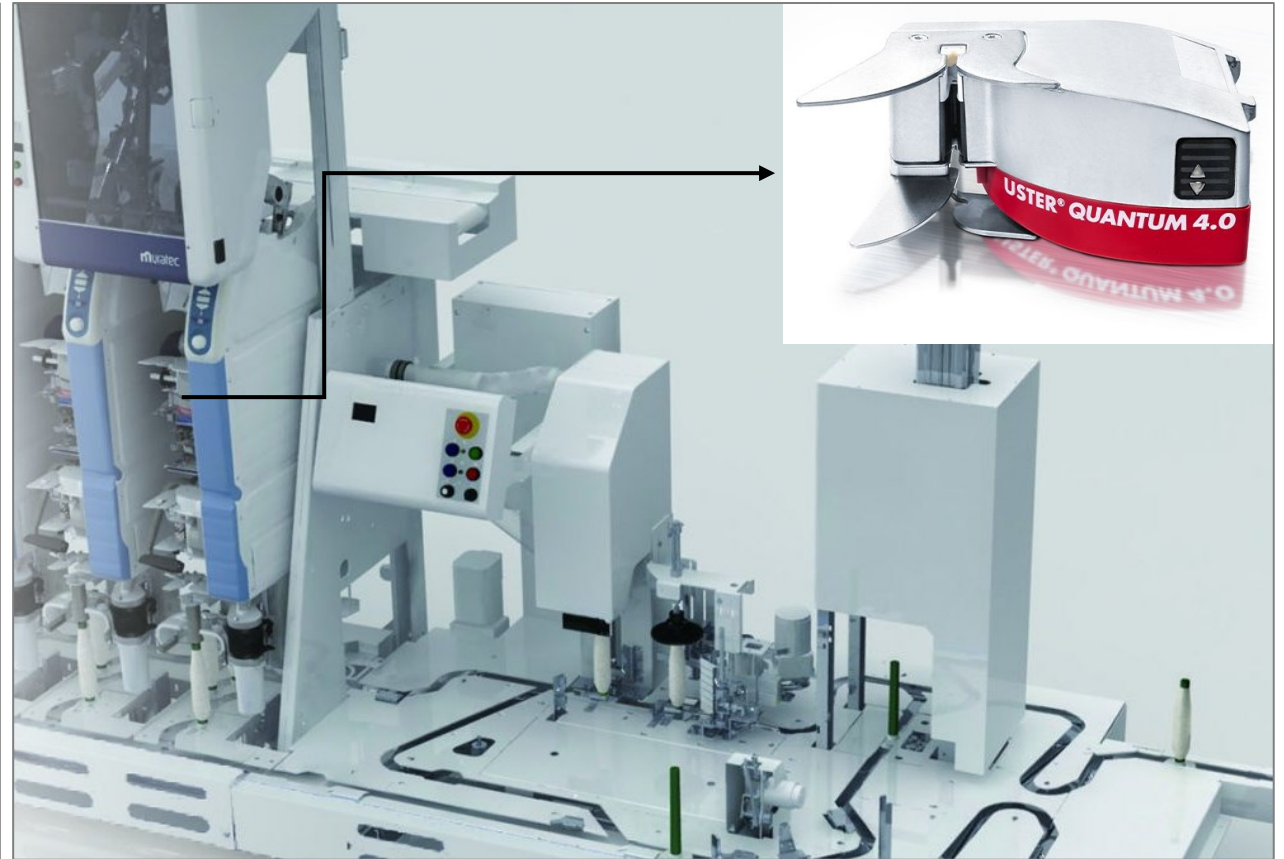
- Feed Material – **Draw frame sliver**
- Machine Components – **Drafting arrangement, spindle, flyer**
- Operations – **Attenuation of draw frame sliver, twisting, winding**

Ring Frame



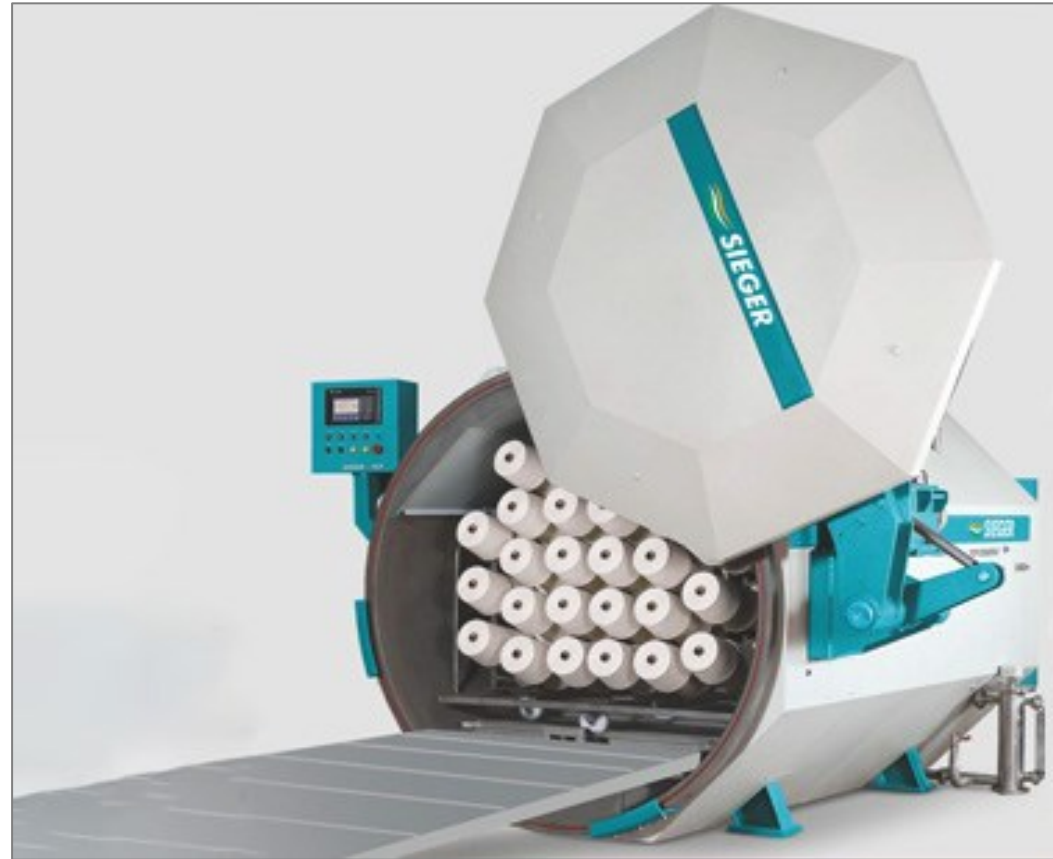
- Feed Material – **Roving**
- Machine Components – **Drafting arrangement, spindle, ring, traveller, compact attachment**
- Operations – **Drafting of roving to the final count, twisting to impart strength, winding**

Link Coner



- Feed Material – **Yarn on Cops**
- Machine Components – **Yarn clearer (Uster quantum), winder**
- Operations – **Yarn Clearing, winding**

Yarn Conditioning Machine



- Feed Material – **Yarn**
- Machine Components – **Yarn conditioning chamber**
- Operations – **Yarn Conditioning**

Packing Machines



Automatic Carton Packing Machine



Pallet Packing Machine

- Feed Material – **Cones**
- Machine Components – **Conveyor, auto packer for cone & carton**
- Operations – **Packing from cone to carton packing**

- Feed Material – **Cartons**
- Machine Components – **Pallet packer**
- Operations – **Pallet Packing**



Proposed Automation

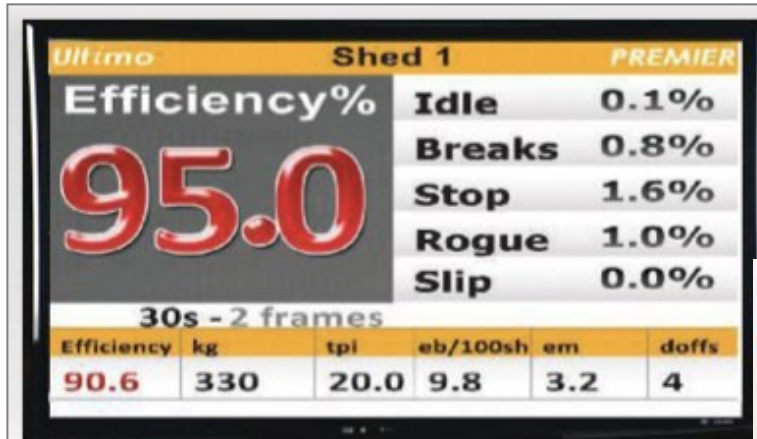


Bobbin Transport System (BTS)



- Improvement in quality due to lack of manual material handling
- No material mix-up
- Effective house keeping
- Traceability for material for each spindle in case of quality issues

Individual Spindle Monitoring



Large Display in Department



App Based control along with fixed station

- Optimization of working environment due to visual indications of occurrence of yarn breaks
- Continuous production monitoring of each spindle which can help in identifying issues and prevent downtime
- Analysis of production, efficiency, end breakage rate, stops, rouge spindles, etc. for each ring frame
- Monitoring of Quality deviations

Cone Transport System (CTS)



- Increased productivity and improved quality
- No material mix-up
- Traceability of the product

Laboratory Equipment

Key Laboratory Equipment

- Fiber Testing
 - Uster AFIS (Advanced Fiber Information System)
 - Uster HVI (High Volume Instrument)
- Uster Evenness Tester – Roving & Yarn
- Uster Tensile Strength Tester
- Uster Classimat
- Wrap Reel
- Yarn Twist Tester
- Lea Strength Tester

AFIS – Fiber Testing



Evenness Tester



HVI – Fiber Testing



Tensile Strength Tester



Solar Power

- Solar power is used in spinning mills to a limited extent due to its capital intensive nature (cost of equipment and land) as depicted in table below:

	Standard Norms*	Values for 30,000 spinning mill
Installed capacity	1 MW	15 MW
Daily Units generated	4,000	Approx. 60,000 (peak requirement)
Land required	4 to 6 Acres	60 to 90 Acres
Capital expenditure	US\$ 0.7 to 1 mn.	US\$ 11 to 15 mn.

*The norms would vary to some extent depending on the location, type of solar tech and offline/online grid arrangement

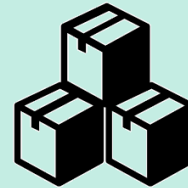
- Most of the mills utilize some part of their roofs to set-up small size solar plant to run light systems and maybe few small, non-critical machinery.
- In the proposed spinning mill, we can cover approx. 4,000 sq. m.(25-30% of the roof) to set-up approx. 0.2 MW solar installation which will generate 800 units/day and cost approx. US\$ 0.14 to 0.20 mn.

Key Risks and Mitigation Strategies



Machinery Related

- Inventory of critical spares, recruitment of technical experts, and training of staff will be crucial for maintenance of machinery during breakdown.



Inventory Related

- Efficient inventory management will be important as the polyester fiber and all consumables are imported.



Manpower Skill Related

- In the initial phase of the project, technical expertise is required which will be catered by expats. Over the due course of time the technical know-how will be transferred to the local manpower.



Business Plan Details



Key Assumptions – 01/02

Particular	UOM	Value
No. of Working Days	Days/year	350
No. of Shifts	Shifts/day	3
Spares and Consumables	US cents/spindle/shift	0.50
Repair and maintenance	US cents/spindle/shift	0.50
Repair and maintenance: Building & others	% of Sales	0.3%
Other Selling & Overhead Expenses	% of Sales	2.5%
Power Consumption	Units/Kg of yarn	2.9
Power Cost	USD/Unit	0.07
Raw Material Cost		
Cotton - Landed Prices	USD/Kg	2.24
Polyester- 1.4SD - FOB Prices	USD/Kg	1.30
Yarn Cost		
24s 100% Cotton Carded Yarn	USD/Kg	3.17
30s 100% Cotton Carded Yarn	USD/Kg	3.29
24s 52/48 Polyester Cotton Carded Yarn	USD/Kg	2.74
30s 52/48 Polyester Cotton Carded Yarn	USD/Kg	2.80

Key Assumptions – 02/02

Particular	UOM	Value
Working Capital		
Raw Material - Cotton	Days	30
Raw Material - Polyester	Days	90
Store & Spares	Days	120
Packing Material	Days	120
Work In Progress	Days	7
Finished Goods	Days	15
Creditors	Days	30
Debtors	Days	30
Raw Material - Cotton	Days	30
Raw Material - Polyester	Days	90
Margin Money for Working Capital	%	25%

Project Cost

Values in USD Million

Project Cost	Value	Share
Land & Site Development	-	
Building	6.59	25%
Plant & Machinery	11.92	45%
Miscellaneous Fixed Asset	4.26	16%
Preliminary and Pre-Operative Expenses	1.19	4%
Provision for Contingencies	2.28	9%
Margin Money for Working Capital	0.51	2%
Project Cost	26.74	

Means of Finance	Share	Value
Equity	10%	2.67
Term Loan	20%	5.35
Grant	70%	18.72
Total		26.74

Key Machinery and Cost

Machine Name	Machine Supplier	Model	No. of machine	Cost of Machine (USD/Unit)	Total Cost (USD/Million)
Blow Room					
Cotton - 700 Kgs	Truetzschler	-	1	1,90,000	0.19
Polyester - 400 Kgs	Truetzschler	-	1	1,60,000	0.16
Card	Truetzschler	TC19I	17	75,000	1.28
Levelling Drawframe					
Cotton	Truetzschler	TD10	3	50,000	0.15
Polyester	Truetzschler	TD10	2	50,000	0.1
Blending Drawframe	Truetzschler	TD9T	6	55,000	0.33
Breaker Drawframe	Truetzschler	TD9T			
Finisher Drawframe	Truetzschler	TD10	7	50,000	0.35
Speed Frame	Toyota/KTTM	FL200	7	1,35,000	0.95
Ring Frame					
100% Cotton	Toyota/KTTM	RX300	8	1,40,000	1.12
PC Blend	Toyota/KTTM	RX300	9	1,25,000	1.13
Winding - Link Coner	Murata	Process Coner II QPRO EX	648 drums	5,000	3.24
Individual Spindle Monitoring System	Premier	Ultimo	1 on each spindle	5	0.15
Bobbin Transport System	Seiger	-	1 on each spindle	20	0.59
Cone Transport System	Seiger	-	1 on each spindle	20	0.59
Compact Attachment	Rieter	-	1 on each spindle	25	0.73
Yarn Conditioning Machine with accessories	Seiger	YCP 1250	1	50,000	0.05
Total cost including packing, freight, commissioning, insurance charges, etc. and infrastructure tax					11.92

Profit and Loss Statement

Values in USD Million

Particular	Y-1	Y-2	Y-3	Y-4	Y-5	Y-6	Y-7	Y-8	Y-9	Y-10
Sales Revenue	11.6	16.9	23.9	25.4	26.4	27.5	28.6	29.7	30.9	32.1
Waste Realization	0.4	0.6	0.8	0.8	0.9	0.9	0.9	0.9	1	1
Total Revenue	12.0	17.5	24.7	26.2	27.3	28.4	29.5	30.7	31.9	33.1
Expenses										
Raw material	8.6	12.5	17.4	18.3	18.9	19.4	20	20.6	21.2	21.9
Power	0.8	1.1	1.6	1.7	1.7	1.8	1.8	1.9	2	2
Wages	0.5	0.7	1	1	1.1	1.1	1.2	1.2	1.3	1.4
Salaries	0.8	0.8	0.8	0.9	0.9	1	1	1.1	1.1	1.2
Spares and Consumables	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Repair and maintenance	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Other Selling & Overhead Expenses	0.3	0.4	0.6	0.6	0.7	0.7	0.7	0.7	0.8	0.8
Packing Cost	0.2	0.3	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5
Repair and maintenance: Building & others	0.03	0.05	0.07	0.08	0.08	0.08	0.09	0.09	0.09	0.1
Water	0.003	0.003	0.004	0.005	0.005	0.005	0.005	0.005	0.005	0.005
Total Expenses	11.3	16.1	22.2	23.4	24.1	24.9	25.7	26.5	27.3	28.2
EBITDA	0.7	1.5	2.5	2.9	3.2	3.5	3.8	4.2	4.5	4.9
EBITDA %	5.8%	8.3%	10.2%	11.0%	11.6%	12.3%	13.0%	13.6%	14.3%	14.9%
Depreciation	4.1	4.1	4.1	4.1	4.1	0.4	0.4	0.4	0.4	0.4
Total Interest	0.4	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
PBT	-3.8	-3.1	-2.2	-1.9	-1.5	2.5	2.8	3.2	3.6	3.9
PBT %	-29.9%	-16.8%	-8.1%	-6.3%	-5.0%	8.8%	9.6%	10.4%	11.2%	11.9%
Tax Payable	-	-	-	-	-	0.6	0.7	0.8	0.9	1.0
PAT	-3.8	-3.1	-2.2	-1.9	-1.5	1.9	2.1	2.4	2.7	3.0
PAT %	-29.9%	-16.8%	-8.1%	-6.3%	-5.0%	6.6%	7.2%	7.8%	8.4%	8.9%

Key Project Numbers

Financials (5 th Year)	USD Million
Revenue	27.27
EBIDTA	3.17
EBIDTA %	11.6%
Working Capital	4.61
Working Capital (as % of Revenue)	16.9%
Ratios	
Average DSCR	2.59
Project IRR (Post-tax)	16.7%
Equity IRR (Post-tax)	19.5%

Production and Sales

Production (Ton)	Y-1	Y-2	Y-3	Y-4	Y-5	Y-6	Y-7	Y-8	Y-9	Y-10
24s 100% Cotton Carded Yarn	982	1,375	1,866	1,906	1,906	1,906	1,906	1,906	1,906	1,906
30s 100% Cotton Carded Yarn	758	1,062	1,441	1,471	1,471	1,471	1,471	1,471	1,471	1,471
24s 52/48 Polyester Cotton Carded Yarn	1,360	1,903	2,583	2,638	2,638	2,638	2,638	2,638	2,638	2,638
30s 52/48 Polyester Cotton Carded Yarn	818	1,145	1,555	1,587	1,587	1,587	1,587	1,587	1,587	1,587
Total Yarn Production for Exports	3,918	5,486	7,445	7,602	7,602	7,602	7,602	7,602	7,602	7,602

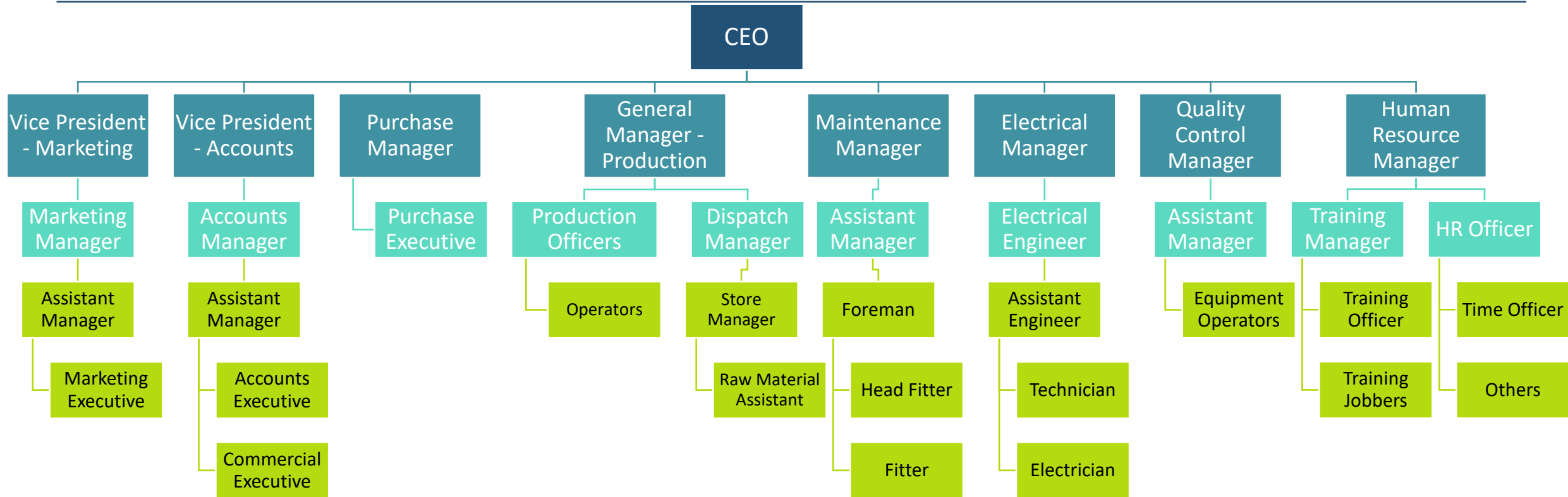
Sales (USD Million)	Y-1	Y-2	Y-3	Y-4	Y-5	Y-6	Y-7	Y-8	Y-9	Y-10
24s 100% Cotton Carded Yarn	3.11	4.53	6.40	6.80	7.07	7.35	7.65	7.95	8.27	8.60
30s 100% Cotton Carded Yarn	2.50	3.64	5.13	5.45	5.67	5.89	6.13	6.37	6.63	6.89
24s 52/48 Polyester Cotton Carded Yarn	3.73	5.43	7.67	8.14	8.47	8.81	9.16	9.52	9.90	10.30
30s 52/48 Polyester Cotton Carded Yarn	2.29	3.34	4.72	5.01	5.21	5.42	5.63	5.86	6.09	6.34
Total Yarn Production for Exports	11.64	16.94	23.91	25.39	26.41	27.47	28.57	29.71	30.90	32.13

Raw Material Consumption and Cost

Raw Material Requirement (Ton)	Y-1	Y-2	Y-3	Y-4	Y-5	Y-6	Y-7	Y-8	Y-9	Y-10
Total Cotton Required	3,179	4,451	6,041	6,168	6,168	6,168	6,168	6,168	6,168	6,168
Total Polyester Required	1,156	1,618	2,196	2,242	2,242	2,242	2,242	2,242	2,242	2,242

Raw Material Cost (USD Million)	Y-1	Y-2	Y-3	Y-4	Y-5	Y-6	Y-7	Y-8	Y-9	Y-10
Total Cotton Required	7.1	10.3	14.4	15.1	15.6	16.0	16.5	17.0	17.5	18.1
Total Polyester Required	1.5	2.2	3.0	3.2	3.3	3.4	3.5	3.6	3.7	3.8
Total Raw Material Cost	8.6	12.5	17.4	18.3	18.9	19.4	20.0	20.6	21.2	21.9

Organization Chart



Employee	Local	Expat*	Total
Worker	245	36	281
Staff	52	13	65
Total Employment	297	49	346

*Details of expat employees is mentioned in annexure



Annexures



Annexure 1: List of Expat Workers and Staff

Workers	Core Operations		Maintenance		Total	Staff	Number of Staff
	Jobber	Reliever	Foreman	Head Fitter			
Preparatory	9	3	2	3	17	CEO	1
Spinning	6	3	1	3	13	Vice President - Marketing	1
Link Coner	3	-	1	1	5	Production Manager	1
Training	1	-	-	-	1	Production Officer	4
Total					36	Training Manager	1
						Maintenance Manager	1
						Quality Control Manager	1
						Electrical Engineer	2
						Assistant Engineer - Electronics	1
						Total	13